

SONBONG, Korea, Democratic People's Republic of

WMO: 470030

Lat: 42.317N

Lon: 130.400E

Elev: 10

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.3	8.1	-15.8	2.3	12.8	-12.0	2.9	14.0	33.3	9.7	29.7	13.0	11.9	320	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.0	81.8	73.5	79.1	72.6	76.9	71.8	76.0	79.3	74.5	77.5	73.0	75.9	6.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	131.4	78.2	73.5	124.6	76.7	72.0	118.3	75.2	39.6	79.1	38.2	77.7	36.8	76.0	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.9	16.7	1.6	88.5	2.8	2.4	-0.3	90.2	-2.0	91.6	-3.5	93.0	-5.5	94.8
			0.2	78.1	2.4	2.9	-1.6	80.2	-3.0	81.9	-4.4	83.5	-6.2	85.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.6	19.9	23.5	32.9	43.9	52.5	60.5	68.1	71.4	62.5	50.8	35.6	24.1
	DBStd	18.33	6.04	6.22	6.07	5.44	5.58	4.99	5.05	3.93	4.87	5.71	7.82	6.61
	HDD50	3732	933	742	530	196	35	0	0	0	0	60	433	803
	HDD65	7451	1398	1162	995	632	388	153	26	3	104	440	882	1268
	CDD50	2131	0	0	0	14	114	316	563	663	375	85	1	0
	CDD65	371	0	0	0	0	1	18	123	201	28	0	0	0
	CDH74	1332	0	0	0	1	9	74	368	794	86	0	0	0
	CDH80	176	0	0	0	0	1	15	43	107	10	0	0	0
Wind	WSAvg	5.5	7.7	7.1	6.3	5.5	5.0	4.2	3.9	3.5	3.7	5.0	6.2	7.6
Precipitation	PrecAvg	31.1	0.3	0.4	0.6	1.8	3.6	4.5	6.8	5.9	3.9	2.0	1.0	0.4
	PrecMax	44.8	1.5	1.8	2.2	6.5	10.0	12.3	17.8	11.5	12.3	7.3	4.4	1.5
	PrecMin	22.2	0.0	0.0	0.0	0.0	0.1	0.6	1.7	0.6	0.4	0.1	0.0	0.0
	PrecStd	6.1	0.4	0.4	0.6	1.3	2.6	2.4	3.7	2.6	3.2	1.7	0.9	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	44.1	54.6	66.7	74.4	81.0	84.5	86.5	81.0	70.1	57.7	44.1
		MCWB	32.6	36.2	42.4	50.6	58.7	66.5	74.3	75.2	68.1	59.2	50.2	38.9
	2%	DB	34.9	39.5	49.2	61.0	69.2	75.3	80.5	82.6	76.1	66.2	54.0	39.8
		MCWB	30.0	33.2	39.6	48.6	57.1	65.3	74.5	74.2	67.0	58.8	47.7	34.7
	5%	DB	32.3	36.4	45.8	57.1	65.3	71.1	77.8	80.3	73.5	63.8	50.9	36.7
		MCWB	28.1	31.7	38.1	47.1	54.8	64.6	73.5	73.6	66.1	57.4	45.2	32.1
	10%	DB	29.6	33.9	42.6	53.3	61.6	68.0	75.6	78.0	71.4	61.5	47.5	33.9
		MCWB	25.8	29.8	36.5	45.4	54.1	63.8	72.3	72.9	65.0	55.6	42.6	29.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.7	37.8	44.4	54.0	62.0	70.5	77.8	79.0	73.0	63.6	52.2	40.3
		MCDB	37.1	41.9	51.8	62.1	69.2	75.8	80.7	82.8	77.4	66.5	56.1	43.1
	2%	WB	31.1	34.7	41.4	51.3	59.3	67.9	75.6	76.7	69.9	61.2	49.1	35.7
		MCDB	34.1	38.0	46.7	58.4	66.2	72.1	78.4	80.3	73.5	64.3	52.8	38.9
	5%	WB	28.4	32.5	39.5	48.7	57.3	65.8	74.0	75.3	68.2	59.1	46.2	32.7
		MCDB	31.7	35.7	43.9	54.2	62.3	69.3	76.9	78.3	71.4	62.5	49.6	35.8
	10%	WB	26.1	30.3	37.6	46.6	55.4	64.1	72.6	74.0	66.6	56.8	43.2	30.4
		MCDB	29.4	33.8	41.8	51.4	59.3	66.9	75.3	76.8	69.9	60.4	47.2	33.6
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.8	13.9	14.1	11.3	8.2	7.1	10.0	15.5	17.1	13.9	13.2
		MCDBR	18.0	17.5	17.9	20.8	19.6	16.0	11.7	14.2	17.5	19.4	19.0	17.2
		MCWBR	14.8	13.5	11.3	11.5	10.0	8.6	7.3	8.5	10.6	14.2	14.5	14.3
	5% WB	MCDBR	18.1	16.8	16.4	17.9	15.8	11.4	9.8	10.3	13.7	17.0	17.9	17.0
		MCWBR	15.5	13.9	11.9	11.7	9.8	7.8	7.1	7.6	10.3	14.5	14.9	14.8
Clear-Sky Solar Irradiance	taub	0.346	0.404	0.478	0.567	0.602	0.596	0.556	0.503	0.446	0.452	0.406	0.358	
	taud	2.128	1.989	1.856	1.711	1.714	1.807	1.978	2.119	2.186	2.027	2.053	2.100	
	Ebn at Noon	241	242	238	227	223	226	233	243	247	224	214	221	
	Edh at Noon	33	44	57	72	73	67	56	47	41	43	36	31	
All-Sky Solar Radiation	RadAvg	708	1004	1358	1545	1574	1534	1393	1437	1359	1053	708	594	
	RadStd	51	64	78	123	170	248	191	119	119	72	75	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page